

Solution Requirements

Date: 15 July 2026

Team ID: XXXXXX

Project Name: Human Development Index (HDI) Prediction using Machine Learning

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can securely log in using email and password. Admin authentication is required for managing datasets and reports.	High
2	Authorization Levels	The system supports different user roles such as Administrator, Researcher, and Policy Maker, each with specific access permissions.	High
3	External Interfaces	The system connects to HDI datasets (CSV/Excel), public development databases, and Machine Learning libraries such as Scikit-learn and Pandas.	High
4	Transaction Processing	The system validates user input, processes development indicators, predicts the HDI category, and stores prediction history.	High
5	Reporting	Generate prediction reports, interactive dashboards, charts, graphs, and downloadable PDF/Excel reports.	High
6	Business Rules	HDI prediction is based on Life Expectancy, Mean Years of Schooling, Expected Years of Schooling, and GNI per capita. The system validates all input values before prediction.	High

7	Compliance to Laws or Regulations	The system protects user information according to applicable data privacy guidelines and ensures secure handling of datasets.	Medium
8	Other	Support comparison of multiple countries, historical trend analysis, and recommendation of development improvement areas.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The prediction process should be fast and responsive.	Prediction generated within 2 seconds
2	Scalability	The system should support increasing numbers of users and larger datasets without performance degradation.	Supports 1000+ concurrent users
3	Security & Data Privacy	User credentials and prediction data must be securely stored and transmitted.	Password encryption and HTTPS communication
4	Reliability & Availability	The system should remain available with minimal downtime.	99.9% uptime
5	Usability & Accessibility	The application should have an intuitive interface and be easy to use on desktop and mobile devices.	User-friendly interface with responsive design
6	Maintainability & Portability	The software should be easy to maintain, update, and deploy on different operating systems and cloud platforms.	Compatible with Windows, Linux, and Cloud environments

Functional Requirement Summary

The Human Development Index (HDI) Prediction System enables users to input development indicators, predict HDI categories using a machine learning model, visualize

results through dashboards, compare countries, and generate analytical reports. The system also provides secure authentication, role-based access, and efficient data processing to support policymakers, researchers, educators, and development organizations.

Non-Functional Requirement Summary

The system is designed to deliver high performance, security, reliability, scalability, and usability. It provides fast prediction responses, secure handling of user data, high system availability, and a responsive user interface while remaining maintainable and portable across multiple platforms. These quality attributes ensure a dependable and efficient HDI prediction solution suitable for real-world applications.